

IBM DATA SCIENCE PROFESSIONAL CERTIFICATE • APPLIED DATA SCIENCE CAPSTONE

Predicting Falcon 9 First-Stage Landing Success

An end-to-end pipeline — data collection, EDA, interactive analytics, and machine-learning prediction of booster recovery.

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Repository: github.com/aashkval/ibm-ds-ps-capstone-spacex

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EXECUTIVE SUMMARY

What we did and what we found

Approach

Collected Falcon 9 launch records from the SpaceX REST API and Wikipedia; wrangled them into a labelled dataset; explored the drivers of landing success with SQL and visualization; built an interactive Folium map and a Plotly Dash dashboard; and trained four classifiers to predict first-stage landing.

Key findings

- Landing success rises with flight experience and varies sharply by orbit and site
- KSC LC-39A is the most successful launch site
- All four models reach ~83% test accuracy; SVM leads on cross-validation
- Landing — hence ~ launch cost — is predictable from public launch attributes

90

Falcon 9 launches

4

ML models compared

0.83

best test accuracy

100

successful missions

Background & problem statement

SpaceX advertises Falcon 9 launches at about **\$62M**, against **\$165M+** for competitors — largely because it recovers and reuses the first stage. Whether that stage lands successfully therefore drives a large share of the cost of a launch.

Problem

Given a planned launch (payload mass, target orbit, launch site, booster version, flight number, reuse history), can we predict whether the Falcon 9 first stage will land?

A reliable prediction lets a competitor estimate the true cost of a launch and inform its bid.

COST OF A LAUNCH

\$62M

SpaceX Falcon 9 (reusable first stage)

\$165M+

Typical competitor (expendable)

The savings come from recovering the booster — so predicting the landing predicts the cost.

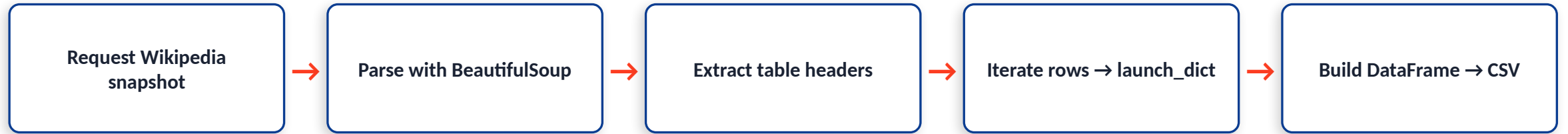
Data collection — SpaceX REST API



- Issued a GET request to the SpaceX v4 API and flattened the JSON with pandas json_normalize.
- Used identifier columns to fetch booster name, payload mass & orbit, launch-site coordinates, and core/landing details.
- Restricted to Falcon 9 and imputed missing PayloadMass with the column mean → 90 launches, 0 nulls.

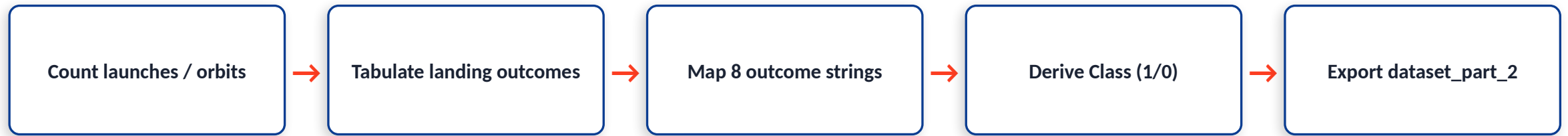
 [notebooks/1 — data collection \(API\)](#)

Data collection — web scraping



- Requested a fixed Wikipedia snapshot (oldid 1027686922) of the Falcon 9 / Falcon Heavy launch list for reproducibility.
- Parsed the launch table with BeautifulSoup, extracting column names from the headers and cell values row by row (with None-guards for missing links).
- Assembled the records into a DataFrame → spacex_web_scraped.csv (121 rows).

Data wrangling & label engineering



- Summarized launches per site and per orbit, then counted the eight textual landing outcomes (True/False/None × Ocean/RTLS/ASDS).
- Created the binary training label Class — 1 when the booster landed successfully, 0 otherwise — by mapping the “bad” outcomes to 0.
- Overall success rate in the dataset ≈ 0.67; result saved as dataset_part_2.csv.

EDA with visualization & feature engineering



- Visualized landing outcome against flight number, payload mass, launch site and orbit (scatter), success rate by orbit (bar), and the success trend by year (line).
- Engineered the model feature matrix: selected predictors and one-hot encoded Orbit, LaunchSite, LandingPad and Serial, casting all columns to float64.
- Result: a fully numeric 90×80 matrix saved as dataset_part_3.csv.

EDA with SQL

Loaded the launch dataset into a SQLite table and answered ten analytical questions with SQL:

- Unique launch sites; first records for sites beginning “CCA”
- Total & average payload mass (NASA CRS; booster F9 v1.1)
- First successful ground-pad landing; boosters with success in 4,000–6,000 kg range
- Counts of successful vs. failed missions; max-payload boosters (subquery)
- 2015 failed drone-ship landings by month; ranked landing outcomes 2010–2017

Interactive analytics — Folium & Dash

Folium map

- Plotted all four launch sites with markers
- Color-coded success / failure via MarkerCluster
- Measured distances to coastline, railway, highway and city using a Haversine helper

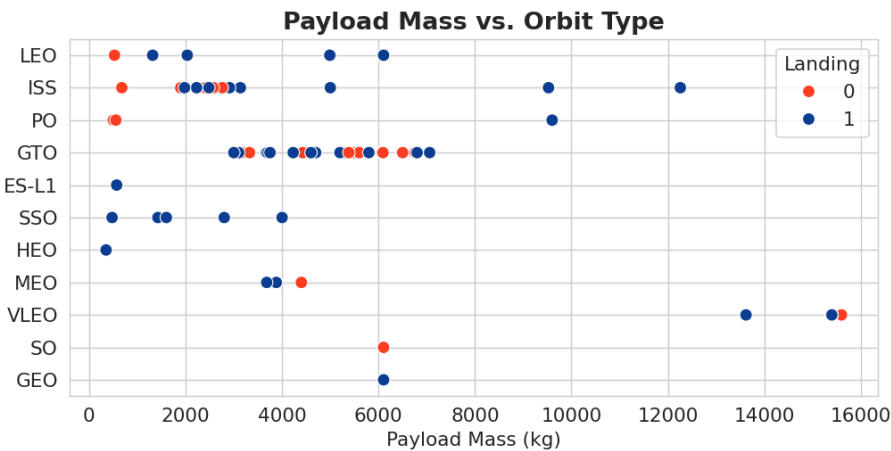
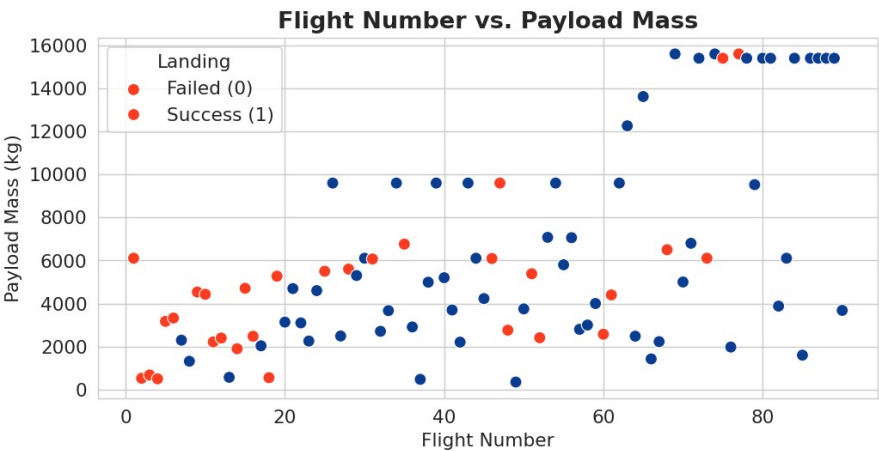
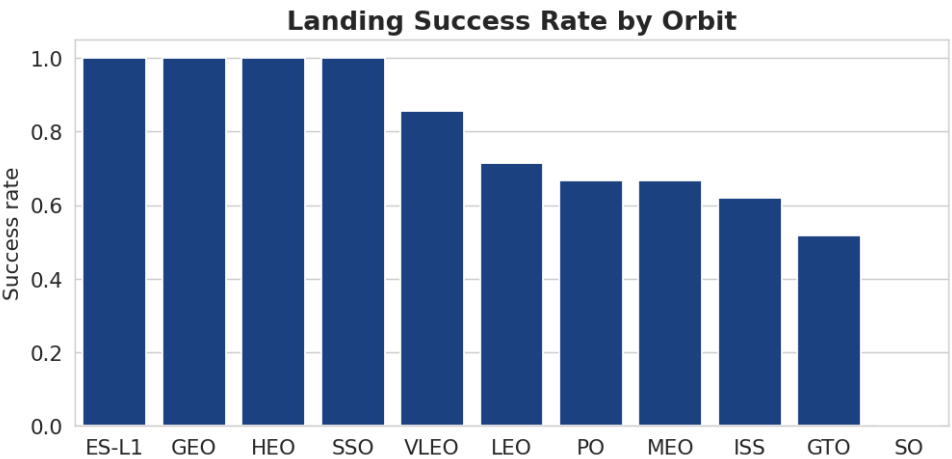
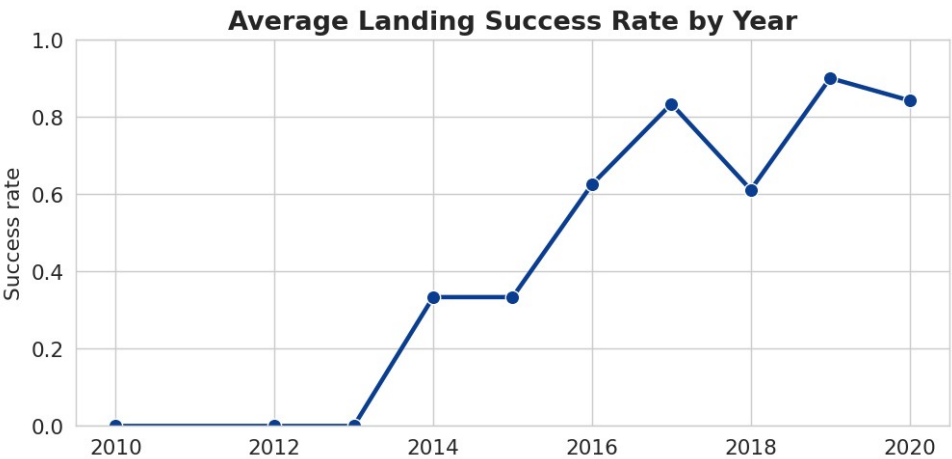
Plotly Dash dashboard

- Dropdown to select all sites or one site
- Pie chart of success share / per-site outcomes
- Payload range slider with an outcome scatter colored by booster version

 [notebooks/6 — Folium map](#)

 [spacex_dash_app.py](#)

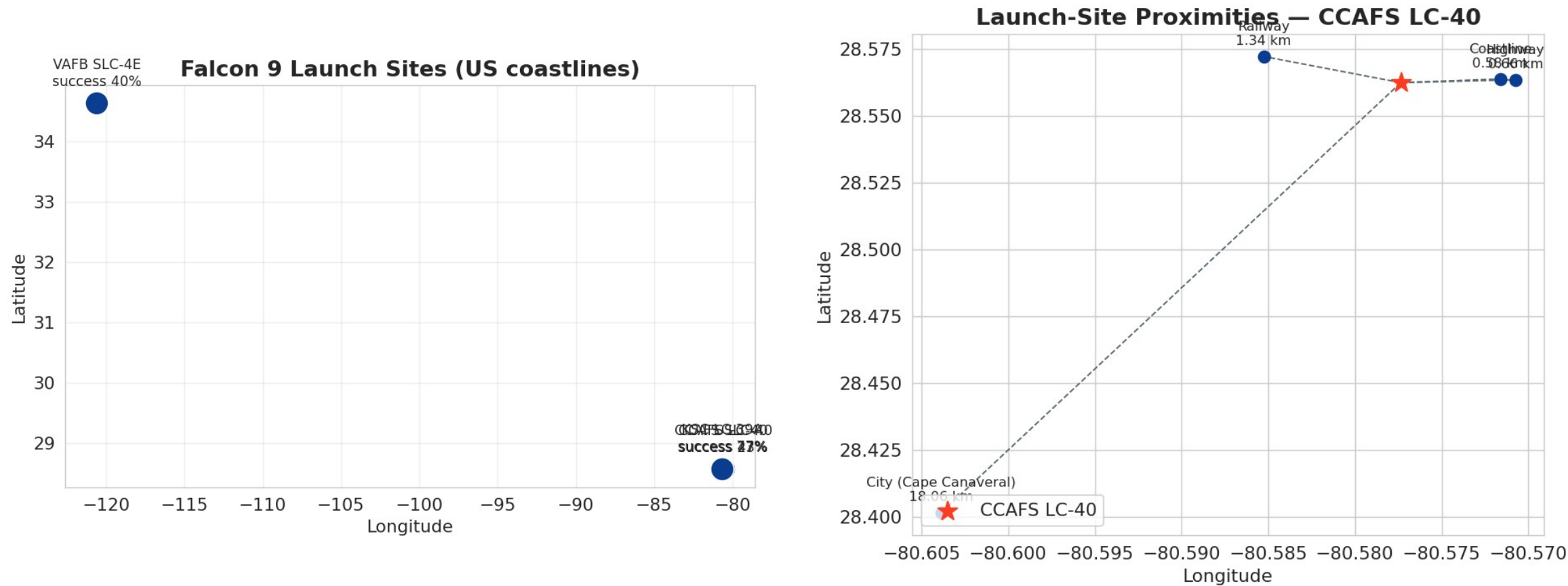
What the charts reveal



Key query findings

Question	Result
Unique launch sites	CCAFS LC-40 • CCAFS SLC-40 • KSC LC-39A • VAFB SLC-4E (4)
Total payload mass — NASA (CRS) customers	45,596 kg
Average payload mass — booster F9 v1.1	2,928.4 kg
First successful ground-pad landing	2015-12-22
Successful vs. failed mission outcomes	100 successful • 1 failure
Boosters carrying the maximum payload (15,600 kg)	Starlink B5 cores (B1048/B1049/B1051/B1060 ...)
Failed drone-ship landings in 2015	2 (January & April, CCAFS LC-40)

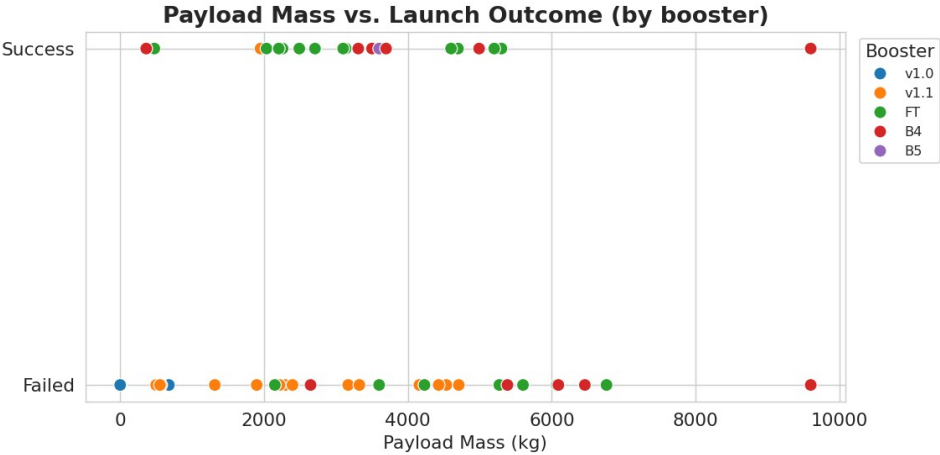
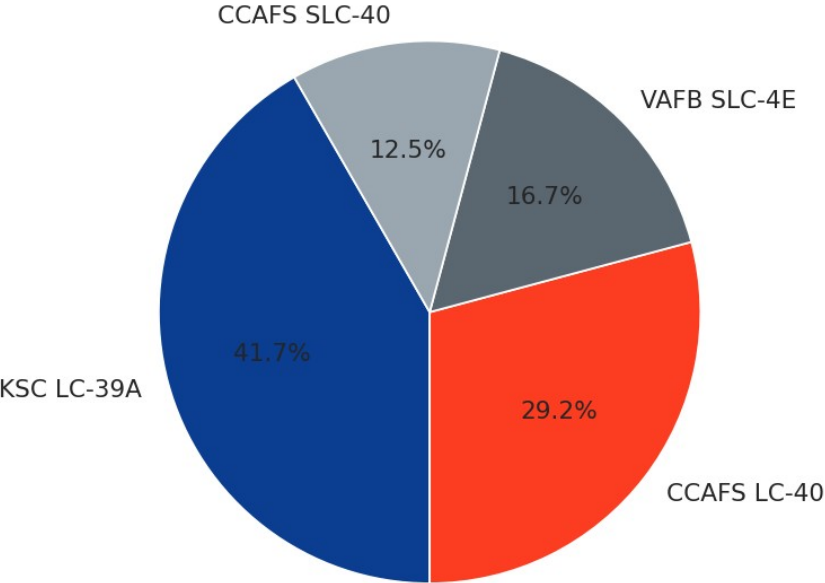
Launch-site geography & proximities



All sites sit on coasts at low latitude. For CCAFS LC-40: coastline 0.58 km, highway 0.66 km, railway 1.34 km, nearest city 18.06 km — close to transport, set back from population.

Dashboard insights

Share of Successful Landings by Site (all sites)



Most successful site

KSC LC-39A

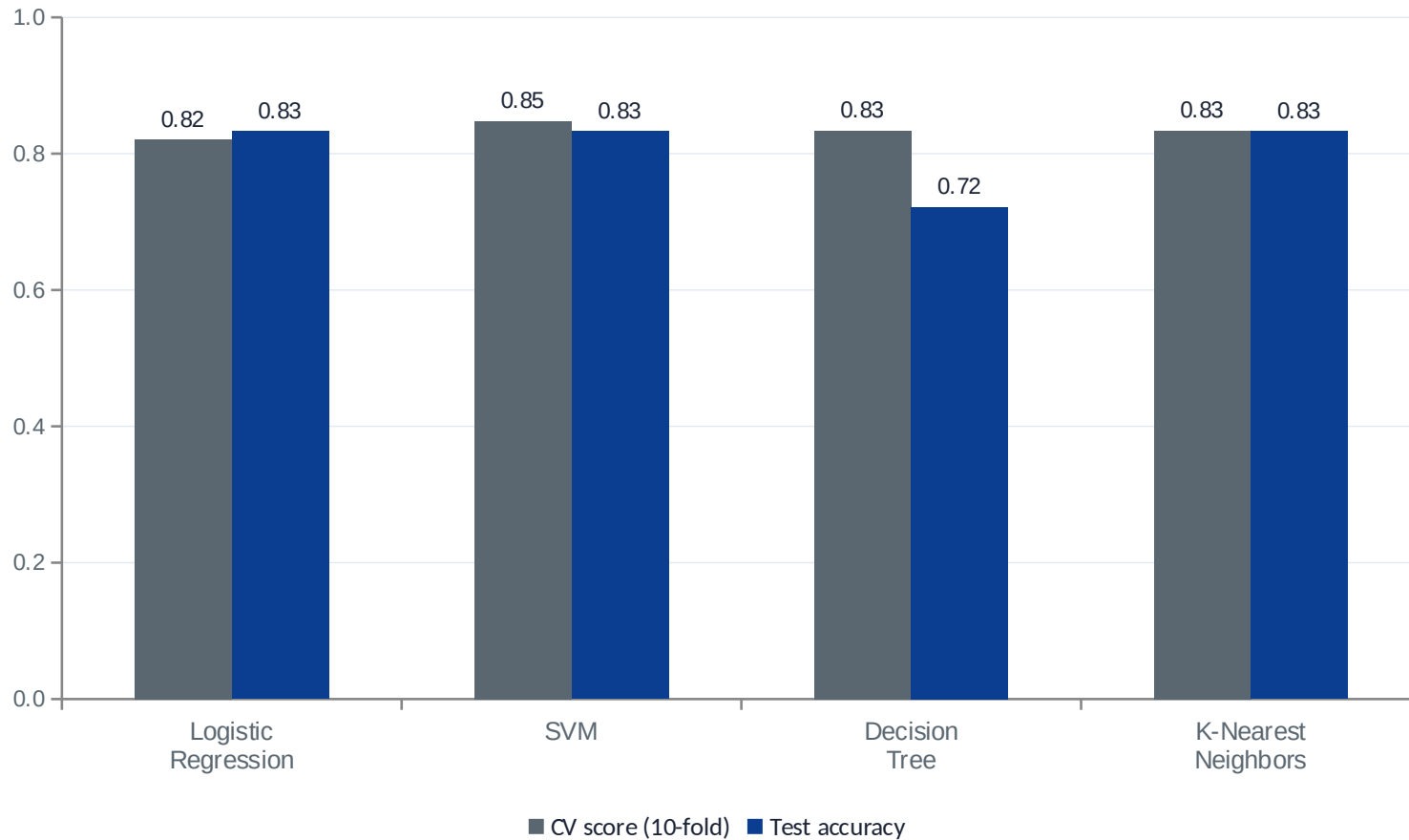
Worst payload band

6,000–8,000 kg → 0%

Best booster

FT ≈ 67% success

Comparing four classifiers



Setup

Standardized 80-feature matrix; 80/20 train-test split (random_state = 2); GridSearchCV with 10-fold CV per model.

Read-out

Three of four models tie at 0.833 on the 18-sample test set. SVM has the highest cross-validation score (0.848).

Best model — confusion matrix

Confusion Matrix — Support Vector Machine (best CV)

True	did not land	land
	3	3
landed	0	12
Predicted		

Support Vector Machine

12 true positives — landings predicted correctly

3 true negatives — non-landings predicted correctly

3 false positives — predicted landing, did not land

0 false negatives — none missed

*The model never misses an actual landing; its only errors are false alarms.
Accuracy on the held-out set is $15 / 18 = 0.833$.*

KEY INSIGHT

Experience drives recovery, not payload size

Across the visual EDA, the dashboard and the four classifiers, the strongest signal for a successful landing is operational maturity — how far into the program and how proven the booster — rather than how heavy the payload is.

A learning curve

Landing success rises steadily with cumulative flight number and year — early failures give way to routine recoveries as the program matures.

Payload is non-monotonic

The 6,000–8,000 kg band lands 0% of the time, while both lighter and some heavier payloads succeed — so payload mass alone is a poor predictor.

The models concur

A linear SVM/LR tops the table (0.833); the axis-aligned decision tree trails (0.722). The signal is largely linearly separable — experience, site, booster — not a payload cutoff.

Implication for a competitor weight booster generation, launch site and flight heritage above payload mass when estimating first-stage recovery — and the ~\$100M cost gap each landing represents.

CONCLUSIONS

What the analysis tells us

1

Recovery is a learning curve Landing success rose steadily year over year and with cumulative flight number — early failures gave way to routine recoveries.

2

Mission profile decides the odds Success depends strongly on orbit type and payload mass; lighter payloads and orbits such as ES-L1, GEO, HEO and SSO land most reliably.

3

Site & booster matter KSC LC-39A is the most successful site and the FT-generation boosters recover most often.

4

Prediction works — with a caveat Four classifiers reach ~83% test accuracy (SVM best on CV). With only 18 test samples the models are effectively tied; more data is needed to separate them.

5

Business takeaway Landing — and therefore the ~\$100M cost gap per launch — can be predicted from public attributes, directly informing a competitor's bid.

Thank you

Full code, notebooks and data:

github.com/aashkval/ibm-ds-ps-capstone-spacex